

CI/CD Observability in Containerized Environments

Project Preparation (PREPD)

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Cybersecurity and Systems Administration

Context and Motivation

Why it matters

CI/CD services: operational infrastructure

Failures: block delivery and incident response

Diagnosis: required across multiple services

Current reality

Logs: scattered across systems

Correlation: weak across services

Decisions: incomplete signals

Problem Statement

Technical constraints

Non-privileged containers: no host-level metrics

Telemetry fragmentation: across CI/CD services

Mixed platforms: Linux and Windows

Organizational constraints

No host privileges: administrative access denied

Security policy: privileged containers forbidden

Diagnosis: slow and manual on-prem

Scope: non-privileged containerized CI/CD on mixed OS

Research Gap

What the literature assumes

Privileged access: host or cloud-managed

Kernel mechanisms: eBPF with elevated capabilities

Sidecar patterns: Kubernetes contexts

Gap in practice

Non-privileged: mixed OS deployments

CI/CD workloads: limited empirical guidance

Ephemeral tasks: post hoc correlation underexplored

Objectives and Research Question

Central question

How can an **integrated observability framework** be designed and evaluated for **privilege-restricted CI/CD** environments?

Objectives

Architecture: container-first without host privileges

Prototype: reproducible images + deployment templates

Evaluation: controlled fault injection + operational metrics

Methodology Overview

Design Science Research Approach:

Phase 1 Problem analysis + PRISMA literature review

Phase 2 Technology assessment and tool selection

Phase 3 Architecture specification + prototype implementation

Phase 4 Evaluation campaign with controlled fault injection

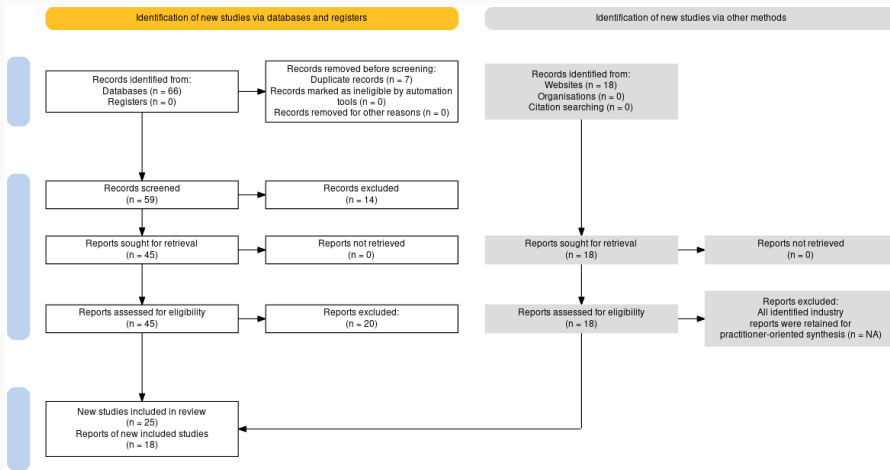
Phase 5 Analysis, thesis writing, defense preparation

Evaluation Strategy:

Quantitative: MTTD, MTTR, resource overhead, throughput

Qualitative: practitioner feedback via structured interviews

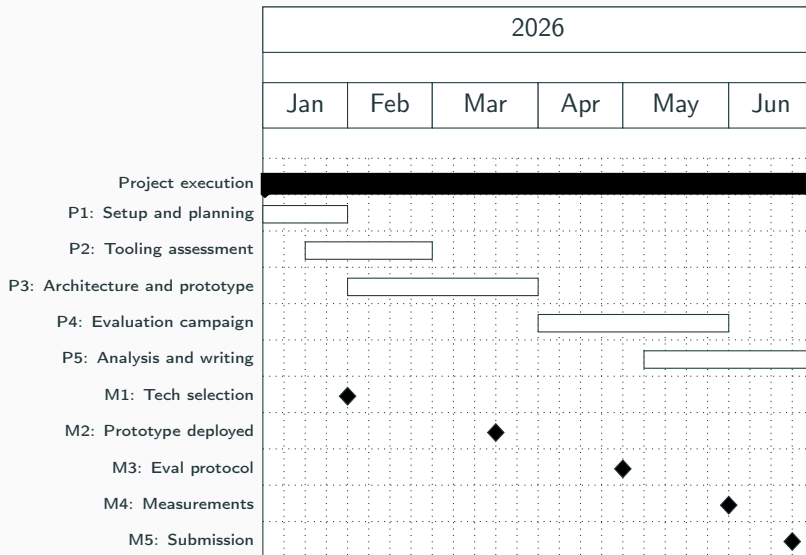
PRISMA Literature Review



PRISMA Summary

- 66** records identified (databases)
- 25** peer-reviewed studies included
- 18** industry sources retained

Project Plan and Timeline



Summary and Next Steps

Established

Problem framing: privilege-restricted CI/CD

PRISMA review: gaps identified

Methodology: execution plan defined

Next phase

Technology selection: constraint-driven

Prototype: architecture + implementation

Evaluation: controlled fault injection

Validation: practitioner feedback

Thank You

Questions?

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